Healthcare Appointment Efficiency Dashboard

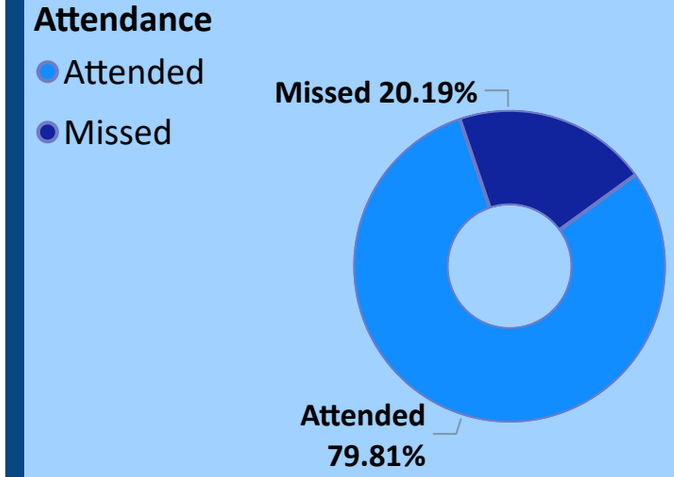
Total Appointments
111K

Attended Appointments
88K

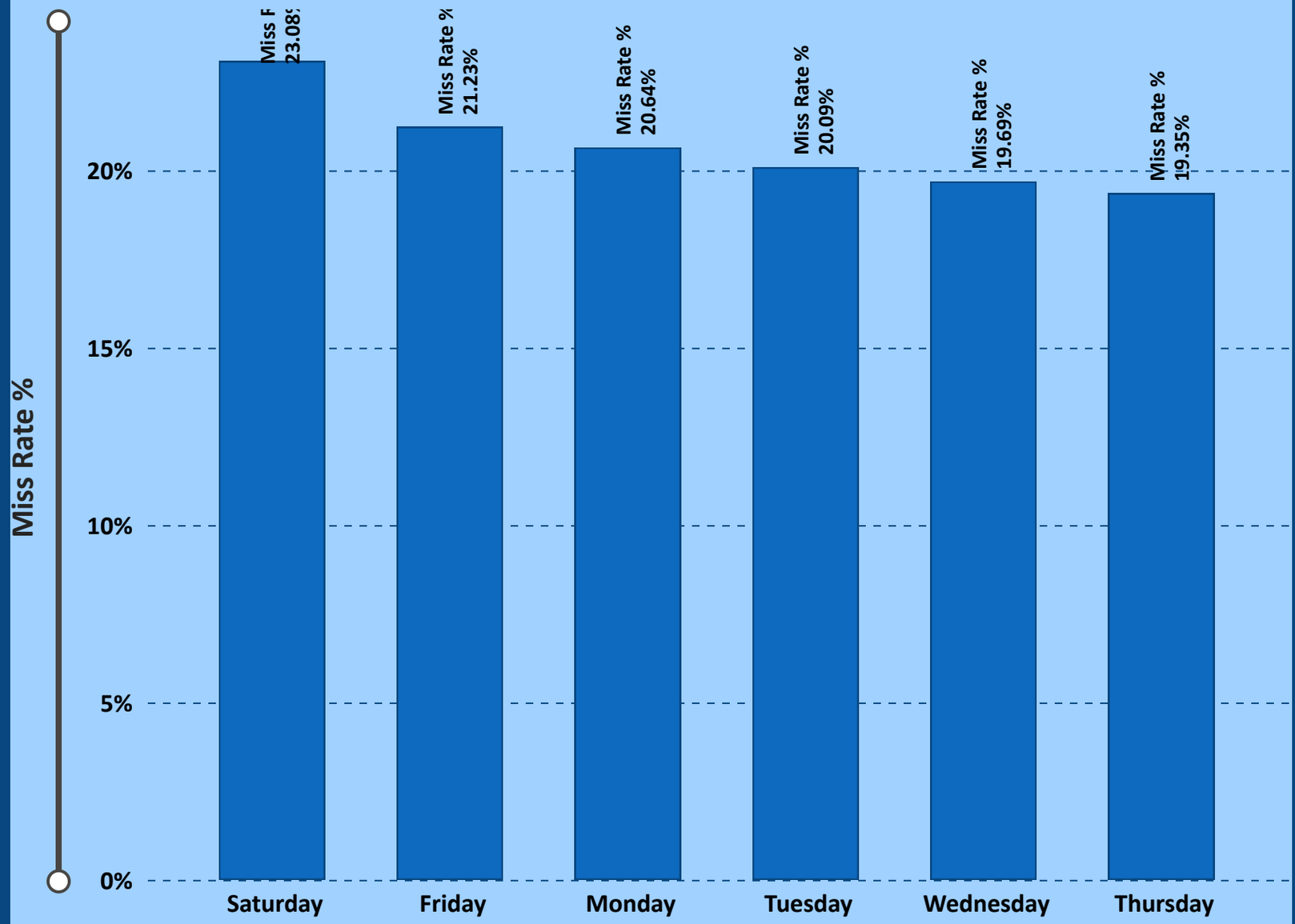
Missed Appointments
22K

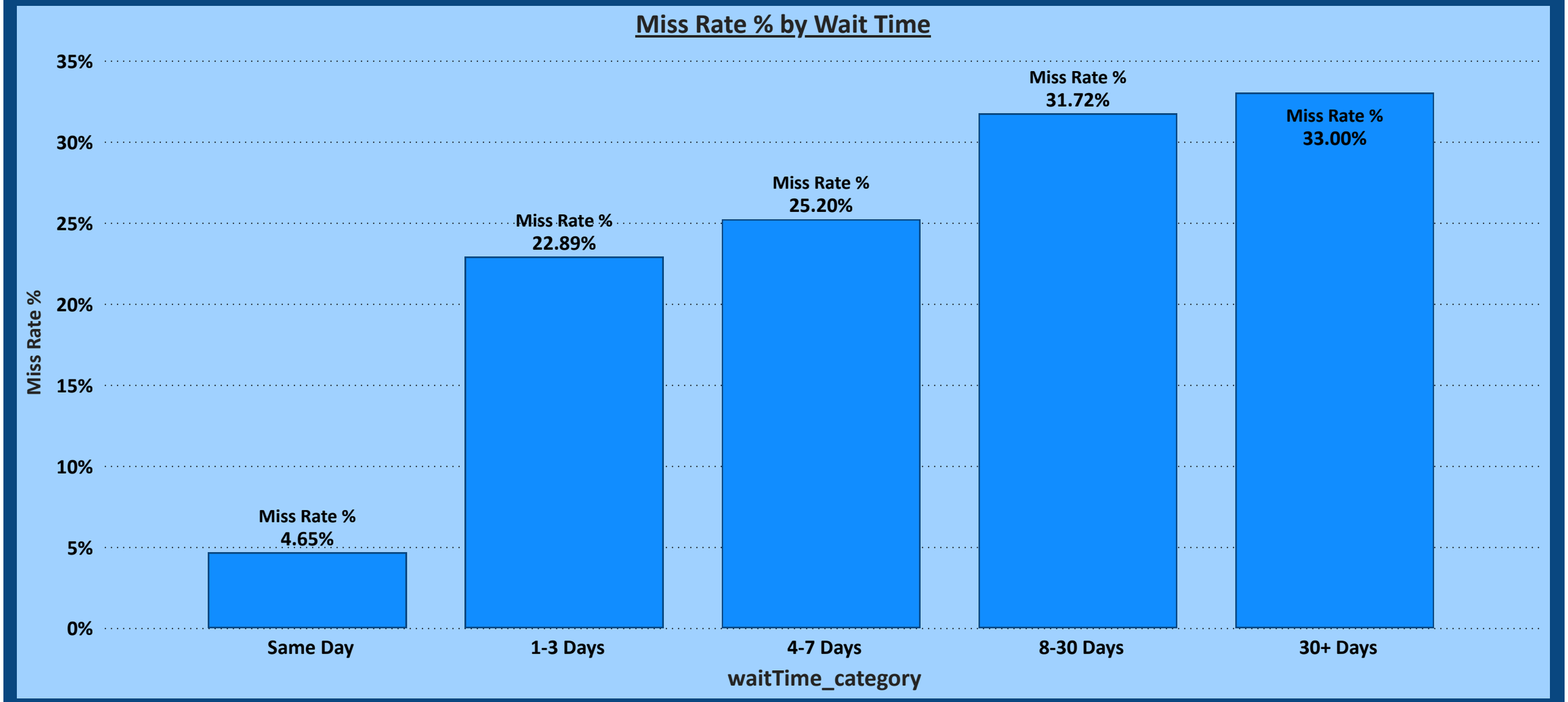
Miss Rate %
20.19%

Appointment Attendance Distribution (Missed vs Attended)

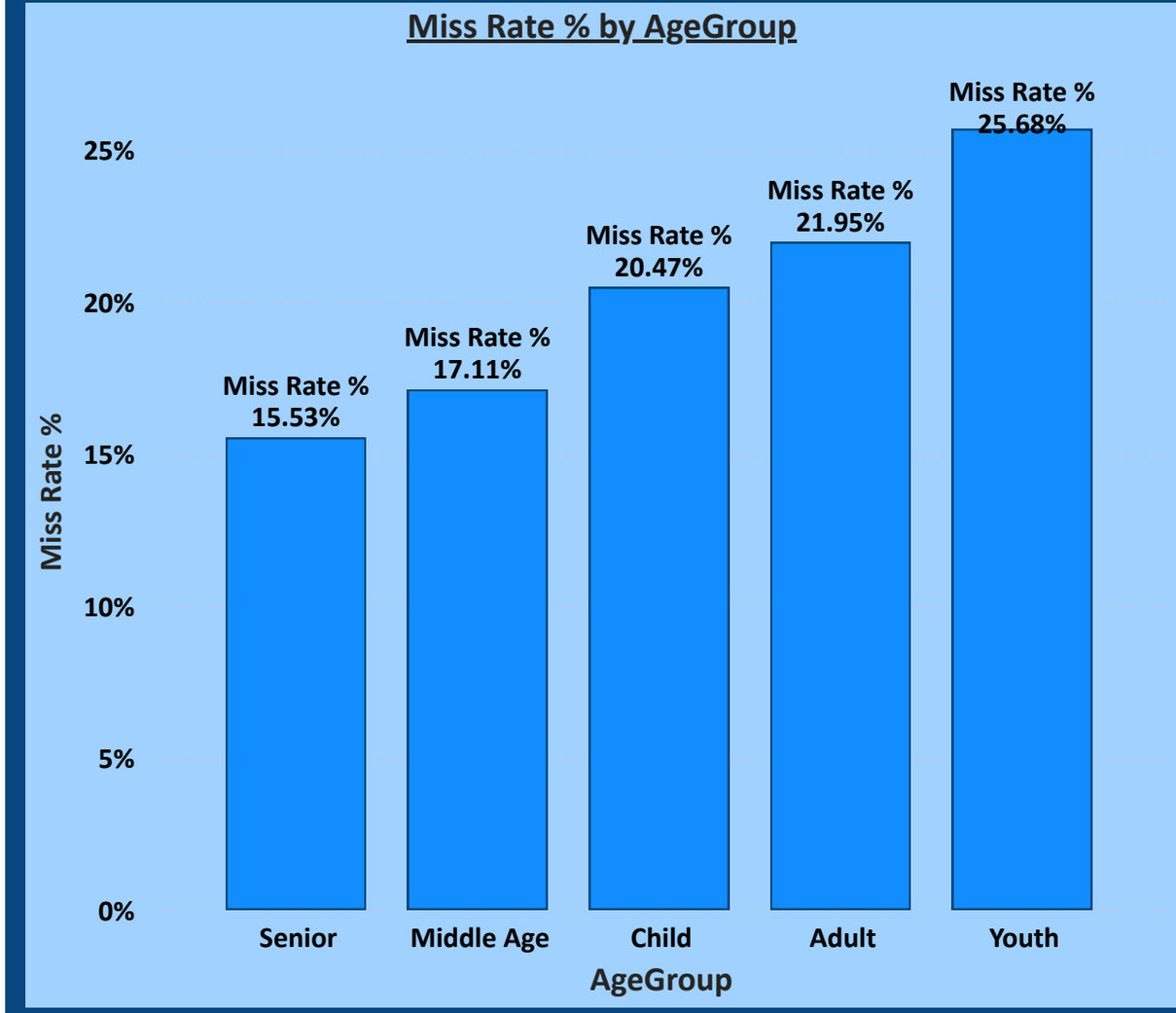


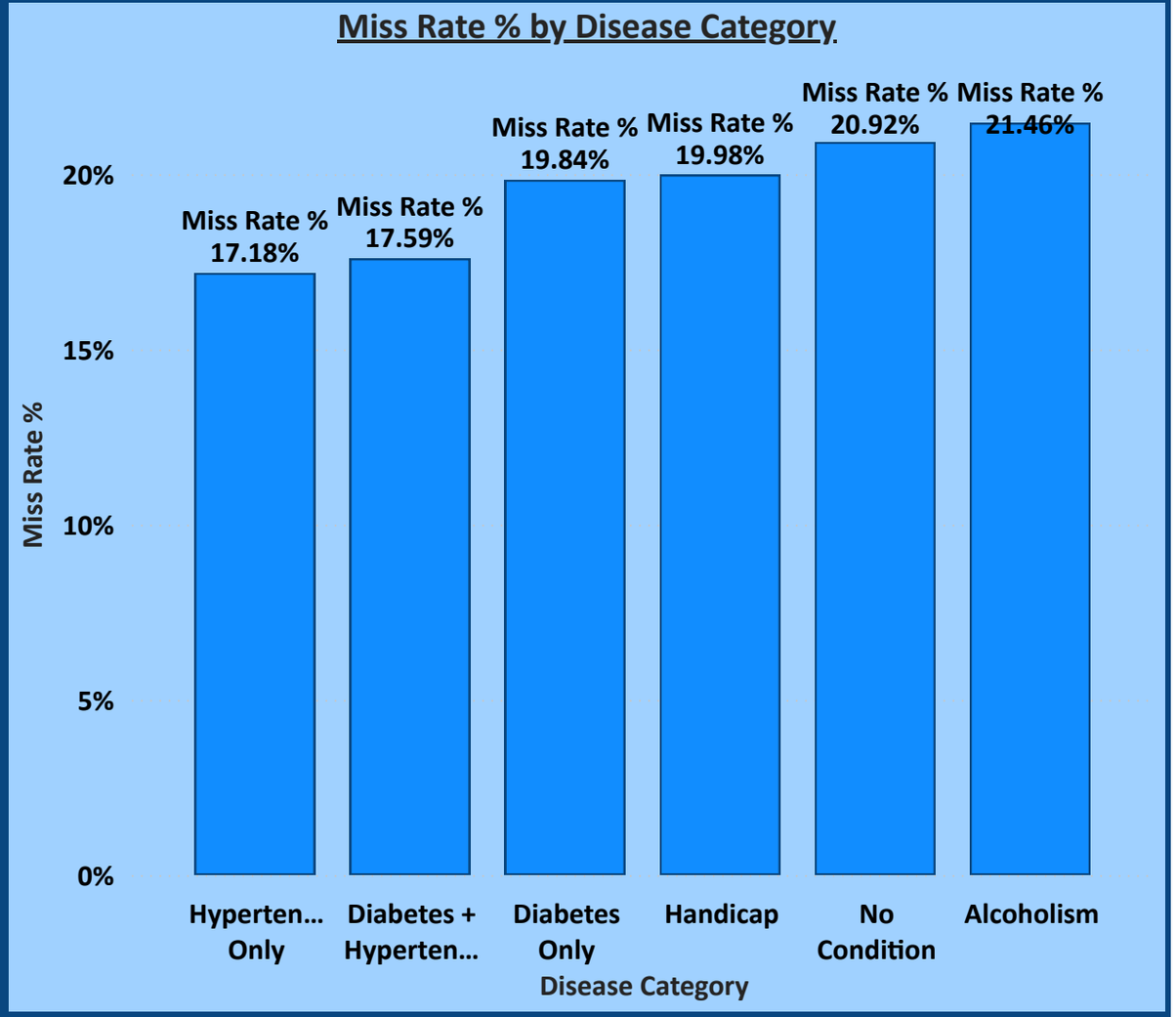
Miss Rate % by WeekDay





- Longer waiting times significantly increase the probability of missed appointments.
- Patients scheduled on the same day have the lowest no-show rate, while appointments scheduled 30+ days in advance have the highest missed rate.

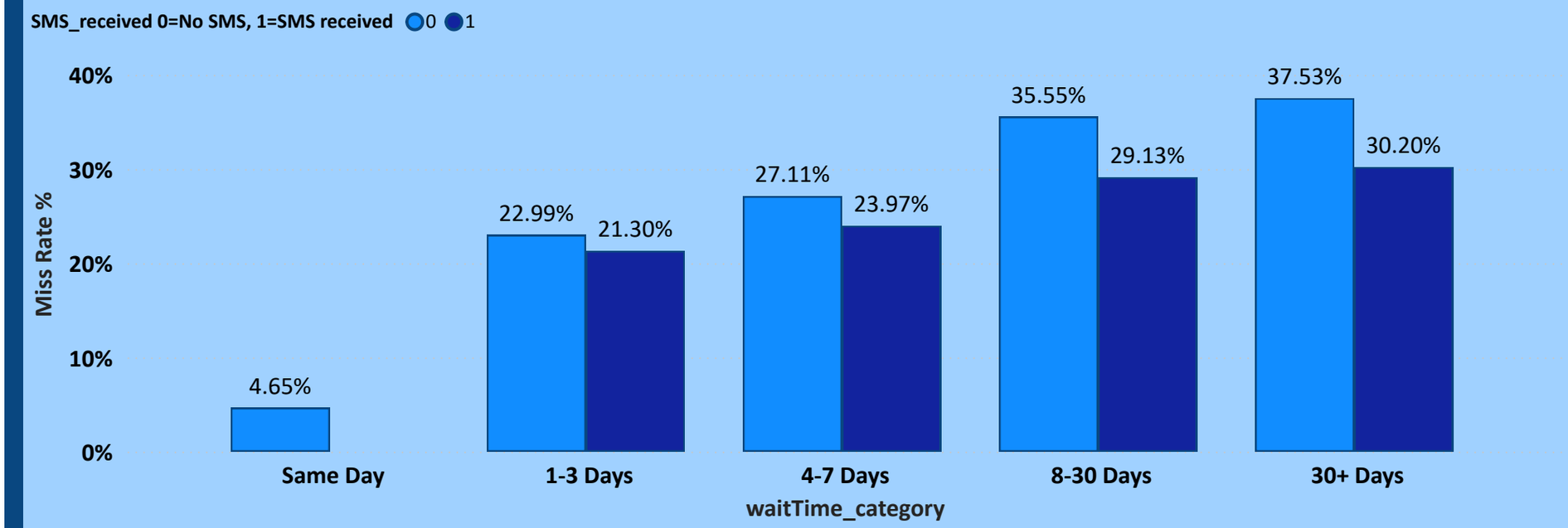




- Younger patients show significantly higher missed appointment rates compared to middle-aged and senior patients. Hospitals may need targeted reminders or scheduling strategies for younger demographics.

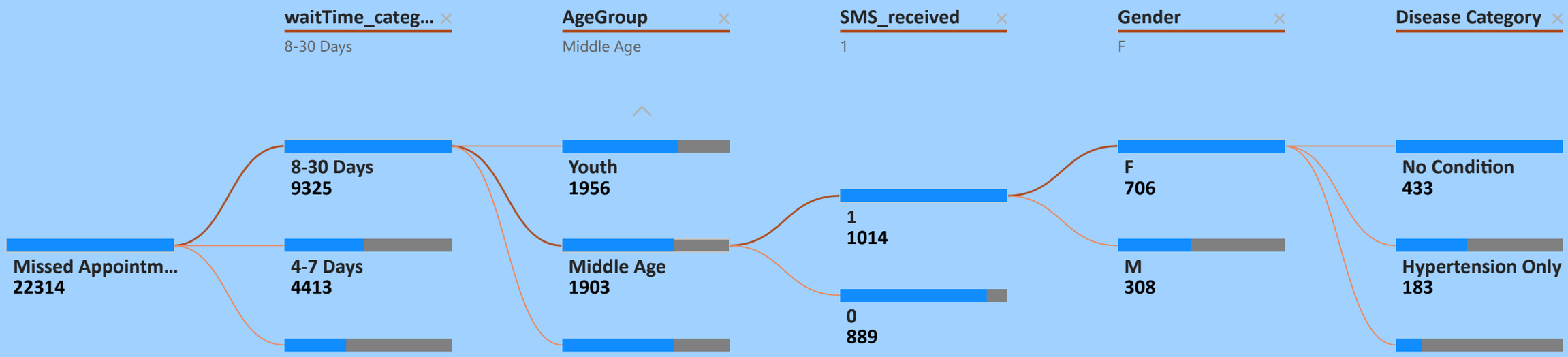
- Patients with alcoholism show the highest missed appointment rate (21.46%). In contrast, patients with chronic diseases such as hypertension and diabetes attend more consistently, likely due to the need for ongoing medical care.

Miss Rate % by waitTime_category and SMS_received



- SMS reminders reduce missed appointments, particularly for appointments scheduled several days or weeks in advance. As waiting time increases, the gap between patients who receive reminders and those who do not becomes significantly larger.

Drivers of Missed Appointments



AI-Driven Analysis

Key influencers Top segments

What influences Attendance to be Missed ?

When...

....the likelihood of Attendance being Missed increases by

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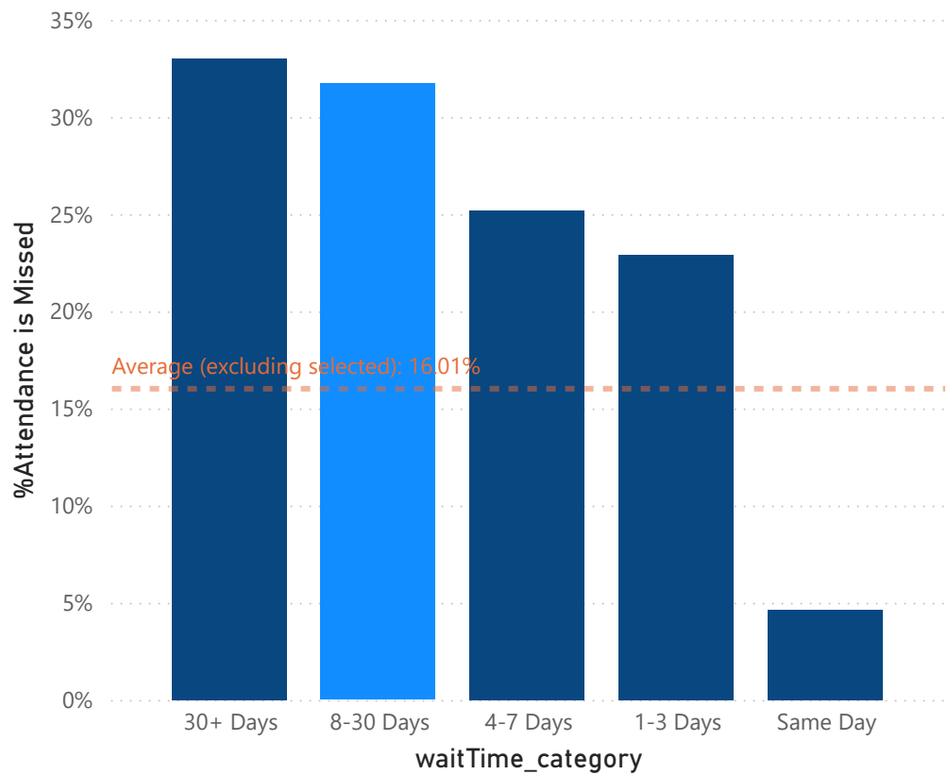
waitTime_category is 8-30 Days	1.91x
waitTime_category is 30+ Days	1.73x
Sum of SMS_received is 1	1.70x
waitTime_category is 4-7 Days	1.39x
AgeGroup is Youth	1.34x
Disease Category is Diabetes Only	1.27x
waitTime_category is 1-3 Days	1.19x

Attendance is more likely to be Missed when waitTime_category is 8-30 Days than otherwise (on average).



Only show values that are influencers

Operational Recommendations

Key Findings:

- Missed appointment rate is **20.19%** of all bookings.
- Patients scheduled **30+ days in advance** have the highest missed rate.
- **SMS reminders reduce missed appointments**, especially for long waiting times.
- **Younger patients and alcoholism patients** show higher no-show behavior.

Recommendation 1

Reduce Long Waiting Times

Appointments scheduled far in advance significantly increase the risk of missed visits.

Recommended action:

- Limit scheduling gaps where possible.
- Introduce dynamic scheduling or overbooking strategies.

Recommendation 2

Strengthen SMS Reminder Systems

SMS reminders consistently lower missed appointment rates.

Recommended action:

- Send reminders **48–72 hours before appointments**.
- Add **same-day reminders** for long waiting periods

Recommendation 3

Focus on High-Risk Patient Segments

Certain patient groups are more likely to miss appointments.

Recommended action:

- Implement targeted reminder campaigns.
- Use predictive alerts for high-risk segments.

Recommendation 4

Use Data Monitoring for Operational Planning

Healthcare systems should continuously monitor appointment behavior.

Recommended action:

- Track missed appointments by **wait time and patient group**.
- Use dashboards to proactively identify risk patterns.